

Figure 2: M1-M5 Model Family Architectural Progression

M1: XGBoost Baseline
Gradient boosted trees
Static summary & hourly features
No self-attention mechanisms



M2: Plain Transformer
3 Transformer encoder layers
Physiological values $v(t)$ only
No observation masks or dt



M3: Time-Aware Transformer
Values $v(t)$ + Masks $m(t)$ + Deltas $dt(t)$
Time2Vec temporal encodings
 $d_model=64$ | Params $\sim 185K$



M4: Organ-Aware Hybrid
Organ system sub-nets (SOFA alignment)
Branching feature routing
Params $\sim 320K$ (Over-parameterized)



M5: Multi-Hybrid / MoE
Mixture-of-Experts routing
Dynamic expert selection
Params $\sim 450K$ (No stat gain, $p=0.068$)